

YUNSOO KIM | University College London

(+44)75-8731-6332 | Email : yunsoo.kim.23@ucl.ac.uk | [Website](#) | [LinkedIn](#)

EDUCATION

University College London, London, UK

PhD in Health Informatics

Sep. 2023 - Present

Thesis: "Foundation Model for Collating Clinical Expert Knowledge: Facilitating Histopathology and Radiology with Multimodal LLMs"

Supervisors: Professor Honghan Wu, Dr Adam Levine, Professor Robert Stewart, Dr Nikolas Pontikos

Imperial College London, London, UK

M.Sc. in Bioinformatics and Theoretical Systems Biology

Oct. 2015 - Sep. 2016

Thesis: "A Clustering-based Short Time Series Analysis Methodology in Omics"

Supervisors: Professor Tim Ebbels, Dr John Pinney, Dr Virginia Fairclough

- [Developed methodology to analyze omics time-series data](#)
- [Developed a novel methodology](#) to infer OTU (Operational Taxonomic Unit) co-occurrence and differential networks
- [Outperformed 8 other inference methods for co-occurrence networks \(the best precision\)](#)

Case Western Reserve University, Cleveland, OH

B.S. with Honors in Systems Biology (Concentration in Bioinformatics, Pre-Health Plan)

Sep. 2012 - May. 2015

Honors Thesis: "Assembly of Marama chloroplast from Next-gen sequencing reads"

Supervisor: Professor Christopher Cullis

RESEARCH INTEREST

Deep Learning in computational biology and health informatics, specifically in the applications of multimodal models for histopathology and radiology.

EMPLOYMENT

LG CHEM, Seoul, South Korea

CTO Platform Technology Artificial Intelligence PJT

Apr. 2020 – Sep. 2023

Specialist. NLP Group Leader. Work Description: Leading natural language processing (NLP) group

Apr. 2020 – Feb. 2022

Professional. NLP Group Leader. Work Description: Leading NLP group

Mar. 2022 – Sep. 2023

- [Developed an artificial reading system using language model \(BERT variants\) to recommend research candidates for researchers](#)
 - Successful cases on different research topics (battery, petrochemical, biopolymer, etc).

LG AI Master Program Pilot at LG AI Research

Sep. 2019 – Jan. 2020

LG Chem representative researcher. Program Description: Solving the challenges using AI and becoming a leader in AI project

- Improved BERT model in biotechnology, especially in Agricultural Biology – Reached SOTA in 7 BioNLP benchmark datasets

CTO Green Bio GBT1 PJT at LG Chem

Mar. 2019 – Mar. 2020

Specialist. Work Description: Developing deep learning models to accelerate biotechnology research

- Developed plant disease recognition and plant disease severity diagnosis with ResNext, EfficientNet ensemble models

CTO Corporate R&D Future Technology GBT PJT at LG Chem

Jul. 2017 – Feb. 2019

Associate. Work Description: Analyzing -omics data to accelerate biotechnology research

- Developed an early allergen and toxin assessment system for the candidate gene
 - [Developed a neural network model for the prediction of protein allergenicity and toxicity with improved performance](#)

Alternative military service, South Korea

Alternative military service at LG Chem, Military Manpower Administration

Aug. 2017 – Aug. 2020

Technical Research Personnel. Description: Alternative military service for the Korean compulsory military service

- Worked both a full-time employee and technical research personnel at LG Chem for the advancement of science in Korea

Torrey Pines Co., Ltd., Cheonan, Korea

Co-Founder (~2023) and CEO. Work Description: Planning a New Brand with microbiome science cosmetics Oct. 2016 – Jul. 2017

- [Bota nouveau Microbiome science hair care cosmetics brand](#)
 - One of the pioneers in hair care cosmetics that introduced microbiome science for hair loss care shampoo

PUBLICATIONS

[Enhancing Human-Computer Interaction in Chest X-ray Analysis using Vision and Language Model with Eye Gaze Patterns](#)

First Author

Conference: MICCAI 2024

[MedExQA: Medical Question Answering Benchmark with Multiple Explanations](#)

First Author

Conference: ACL 2024 BioNLP Workshop

[All you need is continued pretraining of chemistry texts even for molecule captioning](#)

First Author

Conference: ACL 2024 Language Plus Molecule Workshop

[Foundation Model for Biomedical Graphs: Integrating Knowledge Graphs and Protein Structures to Large Language Models](#)

First Author

Conference: ACL 2024 Student Research Workshop

[Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs](#)

First Author

Conference: IJCAI 2024 TAI4H Workshop

[Hallucination Benchmark in Medical Visual Question Answering](#)

Co-first Author

2024

Conference: ICLR 2024 Tiny Paper (Notable Accept)

[Chemical Language Understanding Benchmark](#)

First Author

2023

Conference: ACL2023 Industry Track

[A novel inversion in the chloroplast genome of marama \(*Tylosema esculentum*\)](#)

First Author

2016

Journal: Journal of Experimental Botany

PATENTS

[Literature Chemical Entity Recognition Method and System through Adjusting Sentence Length in Training Data](#)

Main Inventor

2023

Patent Application: KR20240171755A

[AI-based Chemical Entity Recognition System and Method](#)

Main Inventor

2023

Patent Application: KR20240171756A

[Document Classification System and Method Using Entity Group Scores](#)

Main Inventor

2022

Patent Application: KR20240168118A

[Method to build chemical language model and NLP document analysis system and method applying it](#)

Main Inventor

2021

Patent Application: KR20230037335A

[Chemical Named Entity Recognition Method and System through Data Preprocessing](#)

Main Inventor

2021

Patent Application: KR20230030337A

[Crop protection products virtual library construction and browsing system](#)

Co-Inventor

2021

Patent Application: KR20230029084A

[Protein Toxicity Prediction System and Method Using Artificial Neural Network](#)

Main Inventor

2019

Patent Granted: KR102757409B1

HONORS AND AWARDS

FACT Graduate Scholarship

2024

LG Chem On-Spot Incentive

2022

LG Chem DX Good Practice

2021

LG Chem AIT Organization Manager Discretionary Award

2020 - 2021

LG AI IDEathon Digital Transformation Frontier (Applications 1st place; Overall 2nd place)

2021

LG Chem CTO Digital Transformation Best Practice

2021

LG Chem Seoul National University AI Hackathon 1st place

2021

LG Chem DX Best Practice

2020

LG AI Advanced problem solving Best Practice mentor

2020

LG Chem CEO Excellent R&D Project Award

2020

LG Chem Platform Technology On-Spot Incentive

2020

LG Chem Platform Technology Best Memorandum (Research Report)	2020
LG AWARDS (Best Award given by LG Group Chairman)	2020
LG AI Master Best Practice	2019
LG Chem Green Biology Best Memorandum (Research Report)	2019
LG Chem Green Biology On-Spot Incentive	2019
Case Western Reserve University Scholarship	2012 – 2015
Ogelbay Summer Research Grant	2014
Case Western Reserve University Dean's Honor List, 3 semesters	2012 – 2014
Case Western Reserve University Dean's High Honor List, 1 semester	2013
Victor Ridder Scholarship	2012

OTHER EXPERIENCE

Teaching Experience, Workshops, Seminars

MICCAI2025 Tutorial Main Organizer and Speaker	2025
MICCAI2025 Workshop Main Organizer and Speaker	2025
HealTac2025 Tutorial Chair	2024 - 2025
MICCAI2024 Tutorial Main Organizer and Speaker	2024
HealTac2024 Tutorial Main Organizer and Speaker	2024
UCL MSc student co-supervisor for research project	2023 - Present
LG Chem Community of Practice for ChatGPT lead organizer	2023
LG Chem graph neural network model study group lead organizer	2022 – 2023
LG Chem transformer model (NLP) study group lead organizer	2021 – 2023
LG Chem Community of Practice for Systems Metabolic Engineering AI lead organizer	2022
LG AI Advanced problem solving grader, mentor	2020
LG Chem Platform Technology Python Instructor	2020
LG AI weekly study group member, organizer	2019 – 2022
LG Chem Bio Big Data and AI Community of Technology member, organizer	2017 – 2019
Case Western Reserve University Volunteer Peer Tutoring in Biology	2013 – 2014
• Taught two students for biology classes at university-level	

Invited Talks, Presentation

RECOMB Poster Presentation, Lightening Talk	2025
MICCAI Poster Presentation	2024
MICCAI Tutorial Speaker	2024
UCL Institute of Ophthalmology Guest Lecture on LLMs in Medicine	2024
ACL BioNLP, LPM Workshops Poster Presentation	2024
ACL Student Research Workshop Poster Presentation	2024
IJCAI TAI4H Oral Presentation	2024
Lightening Talk at AI in Pathology: Research Meets Reality at the Alan Turing Institute	2024
UCL Computational Pathology Meetings Invited Talk	2024
HealTac PhD Thesis Oral Presentation and Extended Abstract Poster Presentation	2024
ICLR Tiny Paper Poster Presentation	2024
University College London Institute of Health Informatics Talk on Large Language Model Research in Medicine	2024
University of Edinburgh Clinical NLP Group Talk on Large Language Model Research in Medicine	2023
ACL2023 Industry Track Poster Presentation	2023
LG Chem CTO Digital Transformer talks	2022
LG Chem Platform Technology Workshop talks	2022
LG Chem Platform Technology Artificial Intelligence annual meeting speaker	2021
LG Chem Tech Fair talks	2021
LG DX Fair 2021 speaker	2021
LG Chem Digital Transformation talks	2021
LG Chem CTO Digital Transformer talks speaker	2021
LG Chem Platform Technology Workshop talks speaker	2021
LG Chem Digital Transformation Best Practice talks	2020
LG Academy Digital Transformation talks panelist	2020
LG Housys AI Master talks speaker	2019
LG AI Master talks speaker	2019
LG Chem Green Biology quarterly meeting speaker	2019

Journal Reviewing

IEEE Transactions on Medical Imaging, BMC Medical Informatics and Decision Making

Conference Reviewing

ICLR2025, MICCAI2025, HealTac2025